AMIT SHARMA

Website, Github, Linkedin

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PROFILE

- Background in Systems Science (Data Science and Statistics)
- Strong skills coding in Python, SQL, Spark. Worked with industry expert for data extraction and model building on patent documents.
- Excellent communication skills, self-motivated, problem solver. Eager to work in cross-functional team. Tutored students on using Azure HDInsight, Blob Storage and Hive.

SKILLS

Programming	Python (nltk, re, pandas, matplotlib.pyplot, pylab, random, bs4 (BeautifulSoup), requests, typing, spacy), *R (caret, rpart, mice)*, Tableau, Java, MATLAB, HTML5, CSS, Javascript, SQL
Data Analysis	sklearn, pytorch (torch.nn), tensorflow (keras), numpy, gensim, scikit-learn, Gaussian Distribution, Poisson Distribution, SVM, DNN, MLP, CNN, RNN
Framework	Django, Scrapy, Jupyter
Other tools	Tableau, Azure (CD/CI, HDInsight), Git, Spark, Mathematica, MS (Excel, PowerPoint, Word)

EDUCATION

Master of System Science (Sept 2018 – Dec 2019), *University of Ottawa – Department of Computer Science & Electrical Engineering, Ottawa, CA*

Bachelor of Engineering (Nov 2011 – Oct 2015)**,** *Tribhuwan University, Lalitpur, Nepal*

Online Courses

6.86x: Machine Learning with Python-From Linear Models to Deep Learning
BMC: Bloomberg Market Concepts
Microeconomics: The Power of Markets by University of Pennsylvania on coursera
Game theory- Advanced Application, a course from Stanford University, on Coursera

EXPERIENCE

University of Ottawa (September 2018 – December 2019)

Project 1: **Automatic Identification of Inventive Concepts of Patent Documents** (Bell Advanced Research Lab, Ottawa)
Technology/Environment: Python, Google Colaboratory, Jupyter
Supervisor: Professor Tet Hin Yeap
Duration: June 2019 – December 2019

- Completed unsupervised classification of text documents using LDA, NNMF and LSA algorithms.
- Used nltk, gensim, spacy and pyldavis for feature engineering and validating the model to decide which algorithm to adopt. Used pyLDAvis for visualizing the results
- Automated data extraction from a website using python's bs4 and requests libraries. Extracted text is used for training and testing the model. Created classification model with respect to IPC criteria.
- Automated data extraction from xml files of size over 600MB because data extracted through web scrapping was polluted. Used xml library with assumption that quadratic blowup would not occur since trusted authority is providing data.

Project 2: **User Privacy in Data Mining**
Technology/Environment: Python, R
Duration: September 2019 – December 2019

- Synthesized dataset for demonstrating why Anonymization is not a very good method and worked on connecting Gaussian and Laplacian distribution for data randomization on query.

- Demonstrated how secure-multiparty computation or, randomization will help preserve privacy while allowing inference from a dataset, using the technique of Differential Privacy.

Project 3: **Sentiment Classification using Bag of Words Model.**
Technology/Environment: Python, Html, Django
Duration: Jan 2019 – Apr 2019

- Made a binary classification system using nltk, pandas, numpy, tensorflow and Django. Model would be generated based on a set of training data to categorize a specific sentiment class.
- Made the class general enough for generating and processing result from twitter.
- Used different feature engineering methods to increase accuracy from 56% to 83% on test set from deep learning model.

Government of Nepal, Ministry of Health, DOHS LMD, Kathmandu, Nepal – July 2016 to June 2018

Role: Mechanical Engineer

Responsibilities:

- Prepared cost estimates of vehicle repair using current market valuation of the parts for The Secretary of Ministry of Health and Directors and Auditors at Department of Health Services.
- Prepared specification and cost estimates of vehicle to be purchased through competitive bidding.
- Completed size requirement analysis and specification design from scratch for competitive bidding of Refrigeration Truck to transport vaccine.

E&T Pvt Ltd Nepal, Bhaktapur, Nepal – November 2015 to April 2016

Role: CAD and Systems Engineer
Responsibilities:

- Develop solid models of car, tunnels etc. in CATIA for testing simulation software being developed.
- Conduct preliminary study for the flow of air in the intake manifold of engine.

ACTIVITIES

Professional Development Club, Founder and Webmaster, University of Ottawa, Ottawa, CA
System Science Graduate Student Association, Founder and President, University of Ottawa, ON, CA

HOBBIES

Hiking, Jogging, High intensity short duration workouts, Pushups

Reference:

Professor Tet Hin Yeap
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Professor Liam Peyton
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